

Decision Journal: Week 7 Model Choice

Date: 21 July 2026

Decision owner: Shakada Blake

Project: Mercer General Hospital emergency-department triage tool

Status: Recommended to the ED Board, pending approval

Context

- The Week 6 baseline (logistic regression and a depth-5 decision tree) passed Dr Reyes' review, and the ED Board asked whether a more complex model justifies the added cost in compute, interpretability and maintenance.
- ESI-1 patients represent roughly 0.1% of arrivals, so overall accuracy cannot decide the choice; under-triage of a critical patient is the error we most need to avoid.

Alternatives considered

- **Random Forest:** an ensemble of decision trees that vote; robust on tabular clinical data, supports balanced class weighting for rare classes, and exposes global feature importances for governance.
- **Gradient Boosting (HistGradientBoosting):** sequential trees each correcting the last; typically the strongest scorer on tabular data, but slower to train and harder to explain per patient.
- **Small MLP (neural network):** flexible, but the least interpretable option and the hardest to justify to a sceptical clinical audience on a dataset of this size.

Decision

We will carry the **Random Forest** into Phase 3 as the recommended triage model, with logistic regression retained as the documented fallback.

Reasoning

- **ESI-1 recall.** On the axis clinical reviewers weigh most heavily, the Random Forest reached 0.500 against 0.250 for logistic regression and 0.125 for gradient boosting eight of sixteen critical patients identified, versus four and two respectively. Gradient boosting's superior accuracy is irrelevant if it misses fourteen of sixteen.
- **Macro recall and inference time.** It led every model on macro recall at 0.553, indicating the most even performance across all five acuity levels, while inference

stayed negligible at 0.057 ms per prediction. That removes the deployment-cost objection IT Governance raised.

- **Interpretability.** It rates "Medium" on the one-minute test: feature importances are clinically plausible stroke alert, blood pressure, oxygen delivery device and SHAP produced a single-patient explanation in 0.2 seconds. Gradient boosting rates "Low" on the same test.

Things I do not yet know

- Whether the ESI-1 advantage is real or an artefact of sample size the entire result rests on sixteen test patients, a margin of four correct classifications, and needs re-estimating on a larger sample.
- Whether the model holds up on a Caribbean case-mix, and whether it stays reliable when biometric fields are missing the programme's founding concern neither of which has been stress-tested.